An experimental investigation on in-vitro blood clot formation using multi-wavelength photoacoustics

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***Abstract*—Studying blood clots can help us to detect and stage many blood-related diseases, such as atherosclerosis, deep vein thrombosis etc. The modalities like venous ultrasound, MR angiography, and CT angiography are regularly used in clinics to diagnose such disorders, however, the angiography protocols have several limitations. Researchers believe that the photoacoustic (PA) technology can be developed as a noninvasive and contrast agent-free method for detecting blood clots. This is an attempt to understand how PA signal varies at different stages of clotting. We utilized a low-frequency transducer with a center frequency of 2.25 MHz, in combination with optical wavelengths of 532, 700, and 1064 nm, to study the formation of blood clots. The peak-to-peak PA amplitude is increased by ≈21% at 532 nm after 20 min concerning the initial measurement. The same increments are ≈150% and 133% at 700 and 1064 nm, respectively. At the intermediate stages (20 to 50 min), the peak-to-peak values remain almost stable at all three wavelengths. It is apparent from the experimental results that the subtle changes in the PA signals, occurring due to the formation of micro-clots at the early stages, can be captured well by 532 nm. To develop a PA method to detect blood clots, it is necessary to conduct further investigations on the optical parameters that change with blood clotting, namely the absorption coefficient (μ_a), scattering coefficient (μ_s), and scattering anisotropy factor (g). Additionally, a suitable combination of exciting wavelength and ultrasound transducer must be determined.**

***Index Terms*—Photoacoustics (PAs), blood clot, absorption coefficient (μ_a), scattering coefficient (μ_s) and scattering anisotropy factor (g).**

I. Introduction

In the human body, blood plays a critical role in sustaining life by transporting nutrients and oxygen to all living cells. Whole blood contains mainly two types of ingredients, blood plasma and blood cells. In blood plasma, vital amino acids, glucose, fats, enzymes, ions and vitamins are present. Platelets are one of the blood cells, along with Ca^{+2} ions, specifically responsible for blood clot formation and healing of any cut in our body. However, sometimes unwanted clot formation inside blood vessels may risk our lives. To prevent clotting, heparin plays a critical role in making a balance in the human body [1]. In different health conditions, this balance between platelets, Ca^{+2} ion and heparin may get disturbed. Therefore, micro-clots may form which can block the narrow blood vessels of the heart and brain causing stroke, paralysis and even death in the end. There are different types of blood clot-related issues, deep vein thrombosis, and thrombophilias are some of them. It is extremely important to detect and diagnose blood clot formations at early stages.

Researchers all across the globe put their effort into developing numerous types of mechanisms for early detection of blood clots [2]. For example, the prothrombin time (PT) and the activated partial thromboplastin time (APTT) are very common techniques and thrombelastography is another method.

Similarly, photoacoustics (PAs) is a potential alternative technique to assess non-invasive blood clots. During the process of blood coagulation, red blood cells clump together to form larger clusters. In the PA technique, laser pulses incident on the blood sample and the laser energy is mostly absorbed by the clots close to the illuminating surface. Thus, the light beam hardly penetrates into the sample. The penetration depends on the choice of the wavelength probing the sample. In most of the studies, single optical wavelengths were used. Therefore, by choosing an appropriate optical wavelength and ultrasound detector we can accurately detect the position and size of the clot [3]–[6]. Researchers mostly used high-frequency ultrasound detectors ($F_c > 7$ MHz) with higher fractional bandwidths ($FBW > 80\%$). High-frequency transducers are more accurate but also costlier than low-frequency and lower-bandwidth single-element detectors. Our motivation for this study is to investigate the different stages of blood clotting using three different optical radiations and also using a low frequency ($F_c = 2.25$ MHz) single-element ultrasound detector with fractional bandwidth ($BW \approx 70\%$).

In this study, we collected whole blood from a healthy

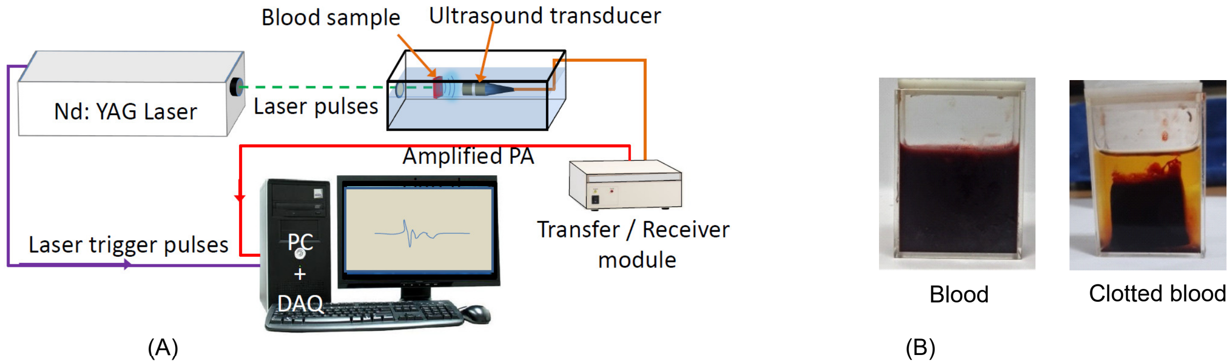

Fig. 1. (A) Schematic diagram of the experiment setup. (B) Blood samples before and after the experiment.

human donor in a vacutainer tube preloaded with an anti-coagulant, ethylene diamine tetra acetic acid (EDTA). The hematocrit level of the blood was 40%. Then $CaCl_2$ solution was introduced to the blood sample to trigger artificial clotting. We employed three different wavelengths 532 nm, 700 nm and 1064 nm as blood has wavelength-dependent optical properties. The choice of wavelengths is very significant for the study. At 532 nm, the Hb absorption coefficient (μ_a) strongly dominates (> 40 times) over any other higher wavelengths for both oxy- and deoxy-Hb. At 700 nm, the absorption coefficient of Hb is very low compared to 532 nm. Further, the absorption coefficients of oxy- and deoxy-Hb greatly differ. Finally, at 1064 nm, the penetration of the photons is much higher as compared to 532 nm and 700 nm and thus this wavelength is suitable for non-invasive assessment [7]. We measured the PA response from the blood sample at different points of time up to 70 min. The most sensitive optical radiation for monitoring in-vitro clot formation is the 532 nm wavelength, compared to 700 and 1064 nm wavelengths.

II. Materials and Methods

A. Sample preparation:

Fresh venous blood of around 8 mL from a healthy human volunteer was collected from the blood bank with the permission of the ethical committee. The blood was divided into three equal volumes and poured into vacutainer tubes prefilled with EDTA at a proportion of 3 g/L to prevent blood coagulation. We took 8 mL of whole blood into a 10 mL cuvette and then placed the sample in the water tank for the PA measurements. The experiments were conducted at a fixed temperature $T = 23^oC$. The whole blood was mixed with 0.6 M $CaCl_2$ solution at $t = 0$ min of our experiment. The experiments were conducted multiple times from $t = 0$ to 70 min and 5 min interval was maintained between two successive measurements.

B. Experimental measurements:

The schematic diagram of the setup used for the PA experiments is shown in Fig. 1(A). We utilized a tunable Nd:YAG solid-state laser system (EKSPLA, NT300 series) that emitted laser pulses with 6 ns pulse width. The laser pulse repetition frequency (PRF) was fixed at 10 Hz. The blood sample was placed inside a glass cuvette (LARK, fluorescent optic glass cuvette). An unfocused single-element ultrasound transducer (Panametric, V320-SU) with a center frequency of 2.25 MHz, 70% fractional bandwidth, and 10 mm diameter captured the PA signals. The detected PA signals were then amplified using a pulser/receiver module (JSR, DPR300). Finally, a data acquisition (DAQ) card (Adlink PCIe 9852) was employed to digitize the PA signals at a sampling frequency of 50 MHz and stored the RF lines each having 5000 data points. The trigger signal from the laser system was utilized to synchronize the DAQ system. For individual blood samples, 200 PA signals were recorded at each optical illumination and then averaged out. The PA measurements were conducted at 532, 700 and 1064 nm optical wavelengths. The blood sample was kept undisturbed till the end of the experiment. The beam energy was measured to be 17.8 mJ/cm^2 at 532 nm, 25 mJ/cm^2 at 700 nm, and 60 mJ/cm^2 at 1064 nm.

III. Results

We measured the PA signals repeatedly with a 5 min interval during clot formation for up to 70 min. For the sake of simplicity, we have shown the PA signal only at one time point i.e., at $t = 0$ min [see, Fig. 2(A), at 532 nm. Multiple pulses with decreasing amplitudes can be seen one after another in the PA signal [see, Fig. 2(A)].

In Fig. 2(B) and (D), the peak of the PA signal detected by the ultrasound transducer reached slightly faster on the detector side as clot formation advanced. The arrows indicate the peaks of the PA signals in the Fig. 2(B)-(D). This is clearly revealed at 532 nm and 1064 nm, but a similar trend is missing at 700 nm wavelength [see, Fig. 2(C)].

In the Fourier space [see Fig. 3(A)-(C)], the major peaks arise around the center frequency of the transducer i.e. $F_c =$ 2.25 MHz, which is consistent with the concept of the center frequency and bandwidth of the particular transducer used in this investigation. The spectral amplitudes at 532 nm gradually increase with clotting time from 0 to 20 min, followed by a decrease at 60 minutes. This pattern corresponds to the PA signals shown in Fig. 2(B). When the amplitudes of the time domain signals increase, the corresponding spectral amplitudes also increase and vice versa. A similar trend can also be observed for wavelengths of 700 nm and 1064 nm [also see Fig. 2(C), (D)].

In the pulse intensity integral (PII) plots [see, Fig. 4], we get three distinct curves for three stages of blood clotting and the spacing between any two curves are different at different wavelengths too. In Fig. 4(A), the spacing between the curves at $t = 0$ min (red) and $t = 20$ min (black) is minimum at 532 nm, compared to those at 700 nm and 1064 nm. By analyzing the PII, it becomes easier to differentiate between the various stages of clot formation i.e., the sudden change that occurs in the PA signal amplitudes.

The plots shown in Fig. 5 portray the time dynamics of peak-to-peak PA amplitude, from $t = 0$ min to $t = 70$ min at 532 nm (right Y-axis), 700 nm (left Y-axis) and 1064 nm (left Y-axis) optical radiations. At 532 nm, the PA signal amplitude is considerably stronger than the PA signals at 700 nm and 1064 nm. The peak-to-peak PA amplitude is increased by ≈21% at 532 nm after 20 min. The same increments are ≈150% and 133% at 700 and 1064 nm after 20 min, respectively. At the intermediate stages (20 to 50 min), the peak-to-peak PA amplitude values remain almost stable at all three wavelengths. The PA signal amplitude is the strongest at the 532 nm wavelength, followed by that of 700 nm and 1064 nm curves.

IV. Discussion and Conclusion

In Fig. 2(B)-(D), there is a phase shift observed in the PA signals at different time points of the blood clot formation. The most probable reason is that the ultrasound waves travel much faster in the clotted samples than the normal blood. Therefore,

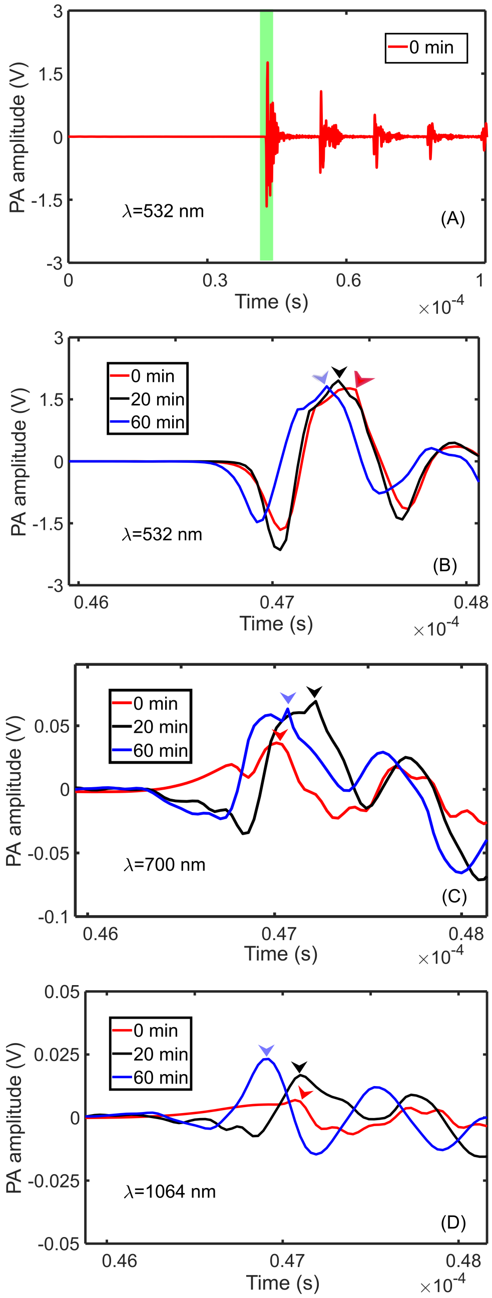

Fig. 2. (A) A representative PA signal measured at 532 nm at t=0. (B)-(D) Zoomed versions of the PA signals at 532, 700 and 1064 nm exciting wavelengths. [The arrows indicate the PA signal peaks.]

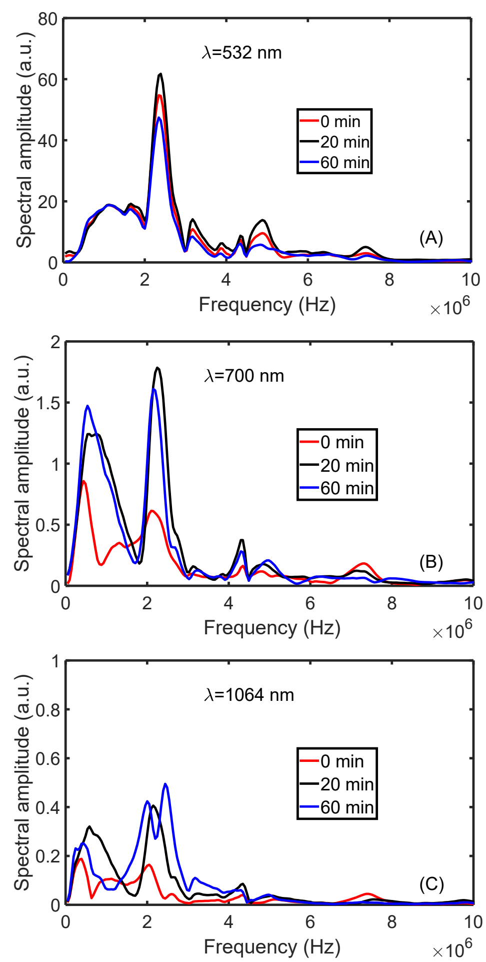

Fig. 3. (A), (B), (C) The Fourier transformation of the PA signals acquired at three different time points (0, 20 and 60 min) using wavelengths of 532 nm, 700 nm and 1064 nm, respectively.

the time taken by the PA waves to reach the detector aperture is different at different clotting stages. In Fig. 5, the PA signal amplitudes are stronger at 532 nm because the absorption coefficient of Hb is much higher (> 40 times) than that of 700 nm and 1064 nm wavelengths. The optical absorption of Hb molecules vis-a-vis PA amplitude at 1064 nm is much smaller than that of 532 nm and 700 nm. Moreover, the variations of the peak-to-peak amplitude line for the 1064 nm (Fig. 5) are not distinctly noticeable because of the choice of the upper limit of the Y-scale (left Y-axis).

It is apparent from the experimental results that the subtle changes in the PA signals, occurring due to the formation of micro-clots at the early stages, the effective morphology of clusters change the scattering coefficient (μ_s) and anisotropy factor (g) of the medium, and these are wavelength dependent [see, Fig. 5]. The change in the anisotropy coefficient alters the fluence distributions of the photons inside the blood samples. The side scattering of photons increases compared to forward scattering as g decreases. The PA signal also increases with decreasing g [7]. The g value dynamically changes (variation is highly wavelength-dependent) during the formation of RBC clusters. Further investigations are required to study how the optical parameters (absorption coefficient, scattering coefficient and scattering anisotropy factor) vary with blood clotting and to develop a PA method, involving a suitable combination of exciting wavelength and ultrasound transducer, for detection of blood clots.

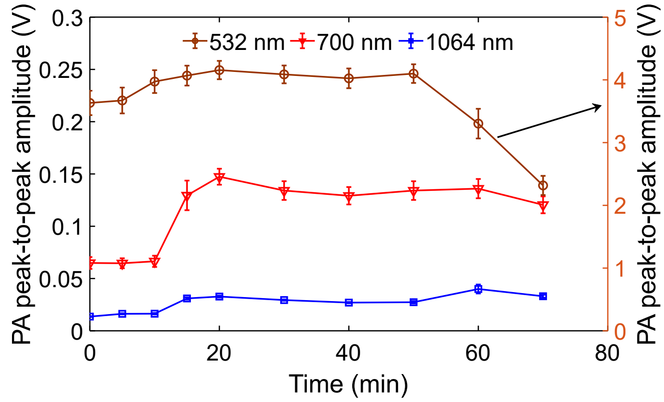

Fig. 5. The dynamical variation of the peak-to-peak PA amplitude at 532, 700 and 1064 nm wavelengths.

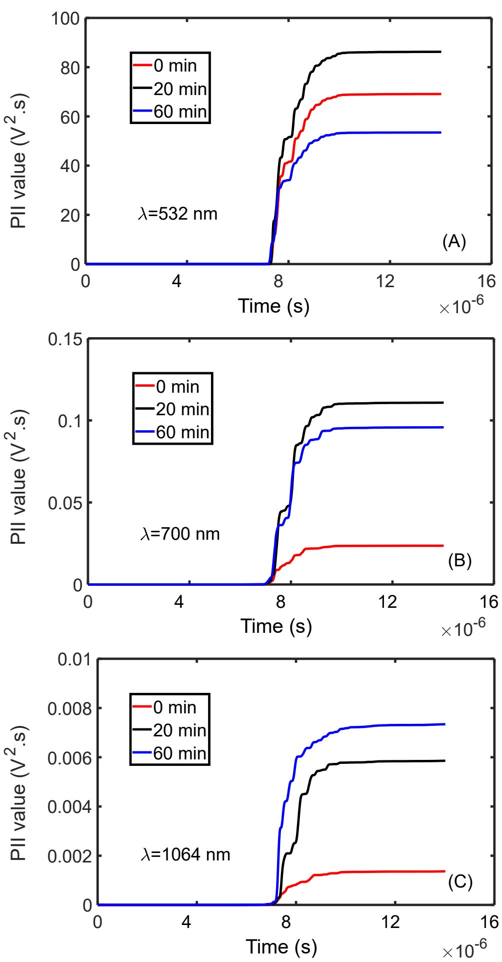

Fig. 4. (A), (B), (C) The plots of the pulse intensity integral (PII) of those PA signals at three different time points (0, 20 and 60 min) using wavelengths of 532 nm, 700 nm and 1064 nm, respectively.

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